

YUWON KIM

Se Young Kim (previous name)

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VR-BASED HUMAN-AI INTERACTION RESEARCHER

A Human-Computer Interaction (HCI) researcher specializing in *Virtual Reality*, *Multimodal Data Fusion*, and *explainable AI* to support people in situated contexts. Dedicated to pioneering innovation in the

My work centers on designing AI that serves diverse stakeholders — clinicians, patients, designers, and everyday users alike — as they pursue distinct goals, combining VR-based behavioral sensing, machine/deep learning, and interpretable AI design. I have applied and stress-tested this approach in one of the most demanding human-AI collaboration settings — clinical decision support — developing VR-based digital biomarker systems and explainable multi-agent AI pipelines through close collaboration with clinical experts (Asan Medical Center, Seoul), engineers at a national research institute (KITECH), and international academic partners (Zurich University of Applied Sciences). I am now looking to extend this foundation to broader situated, embodied, and everyday human-AI interaction contexts.

Dedicated to pioneering innovation in the medical *Artificial Intelligence (AI)* systems sector through advanced digital-based precision medicine system. Concentrating on "**Building Human-centered AI for different medical stakeholders, such as patients and doctors, to accomplish their specific goals,**" leveraging capabilities in medical data analytics and practical application of machine/deep learning techniques. Developed globally competitive Medical AI-HCI technologies through close cooperation with medical experts (Seoul Asan Medical Center), engineers from national research institute (KITECH), and international academic partners (Zurich University of Applied Sciences).

Education

PhD Candidate , Department of Applied Artificial Intelligence Seoul National University of Science and Technology , Seoul, Korea	2022.03.01. ~ 2027.02. (expected)
• Advisor: Prof. Kyoungwon Seo • Thesis: (in progress) An Information Fusion Framework of VR-Based Digital Biomarkers for Early Screening of Mild Cognitive Impairment	

BS , ICT Artificial Intelligence (Interdisciplinary Convergence Double Major) Seoul National University of Science and Technology , Seoul, Korea	2020.03.16. ~ 2022.02.17.
• Advisor: Prof. Kyoungwon Seo, Prof. Ji-Hyeong Han • Thesis: A Study on the Cognitive Load in Stressful Situations Through Eye Tracking Analysis in Virtual Reality Kiosk Ordering Task	

BS , Optometry (Primary Major) Seoul National University of Science and Technology , Seoul, Korea	2016.03.02. ~ 2022.02.17.
• Advisor: Prof. Hyeran Noh • Thesis: Malaprade Reaction-Based Enhancement of Color Vividness in Paper Sensors and Proposal of Efficient Conditions for a Target Shrinkage Ratio	

Journal Articles (More info: [Google Scholar](#))

[J4] Park, B., **Kim, Y.**, Park, J., Choi, H., Kim, S. E., Ryu, H., & Seo, K. (2024). Integrating biomarkers from virtual reality and magnetic resonance imaging for the early detection of mild cognitive impairment using a multimodal learning approach: validation study. *Journal of medical Internet research*, 26, e54538. [SCIE, top 10%]

[J3] Kim, D., **Kim, Y.**, Park, J., Choi, H., Ryu, H., Loeser, M., & Seo, K. (2024). Exploring the relationship between behavioral and neurological impairments due to mild cognitive impairment: Correlation study between virtual kiosk test and EEG-SSVEP. *Sensors*, 24(11), 3543. [SCIE, Q2]

[J2] **Kim, S. Y.**, Park, J., Choi, H., Loeser, M., Ryu, H., & Seo, K. (2023). Digital marker for early screening of mild cognitive impairment through hand and eye movement analysis in virtual reality using machine learning: First validation study. *Journal of medical Internet research*, 25, e48093. [SCIE, top 10%]

[J1] **Kim, S. Y.**, Park, H., Kim, H., Kim, J., & Seo, K. (2022). Technostress causes cognitive overload in high-stress people: Eye tracking analysis in a virtual kiosk test. *Information Processing & Management*, 59(6), 103093. [SCIE, top 10%]

Selected Conference Papers (More info: [Google Scholar](#))

[C4] **Kim, Y.** (2025, April). Beyond Missing Data: A Multimodal Approach Using VR-EEG-MRI (VEEM) Biomarkers for Detecting MCI. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (pp. 1-7). [Research Competition Second-Place Award]

[C3] You, D., **Kim, Y.**, & Seo, K. (2025, April). Tailored Virtual Agent Guidance for Stress Management: Comparing Directive and Non-Directive Approaches by Alexithymia Status. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (pp. 1-8). [acceptance rate: 32%]

[C2] **Kim, Y.**, Park, J., Choi, H., Loeser, M., Ryu, H., & Seo, K. (2024, May). Decoding behavior: Utilizing virtual reality digital marker and machine learning for early detection of mild cognitive impairment. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (pp. 1-8). [acceptance rate: 33%]

[C1] Park, B., **Kim, Y.**, Park, J., Choi, H., Kim, S. E., Ryu, H., & Seo, K. (2024, May). Exploring the Multimodal Integration of VR and MRI Biomarkers for Enhanced Early Detection of Mild Cognitive Impairment. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (pp. 1-8). [acceptance rate: 33%]

Patents

[P2] Seo, K., **Kim, S. Y.** (2025). VR(Virtual Reality) Digital Biomarker based Clinical Decision Support System and Method using Artificial Intelligence for Early Diagnosis of Alzheimer's Disease. *KR10-2025-0022344*. (published 2025.02.17)

[P1] Seo, K., **Kim, S. Y.** (2024). Eye tracking data calibration method and eye tracking data calibration system. *KR10-2024-0048131*. (published 2024.04.15)

Research Grants

NRF, Research Subsidies for Ph.D. Candidates (PI) 25,000k KRW 2025.09.01. ~ 2026.08.31.
(Title: LLM-Based Clinical Decision Support Tool for Alzheimer's Disease via Multimodal Reasoning Integrating VR, MRI, and Neuropsychological Tests)

Awards and Honors

Seoul AI Tech Graduate Scholarship (Seoul Scholarship Foundation, 10,000k KRW)	2025.
Research Competition Second-Place Award (ACM CHI 2025)	2025.

Teaching

Deep Learning (Assistant)	Fall 2022
- Learn deep neural network principles, mathematical foundations, and key models including autoencoders and GANs - Conduct a term project applying deep learning models to address real-world problems	

References

Prof. Kyoungwon Seo

Prof. Sidney Fels

Prof. Seong-Eun Kim

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